

BABU BANARASI DAS **UNIVERSITY**

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SCHOOL OF COMPUTER APPLICATIONS

No SQL Project On

Student Management System Using MongoDB



Submitted to:

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ACKNOWLEDGEMENT

I sincerely express my gratitude to my teacher for giving me the opportunity to work on this NoSQL project titled "Student Management System Using MongoDB".

This project helped me understand the basic concepts of NoSQL databases and learn how MongoDB is used to create, retrieve, update, sort, and delete data. I also learned how different query operators such as greater than, less than, equal to, AND, and OR are used in MongoDB.

I am thankful to my teacher for the guidance and support provided throughout the completion of this project.

INTRODUCTION

MongoDB is a popular NoSQL database management system that stores data in the form of documents. Unlike traditional relational databases, MongoDB does not require data to be stored in fixed rows and columns.

In this project, a Student Management System has been created using MongoDB. The database contains information about 50 students, including their roll number, name, age, marks, and city.

Different MongoDB operations have been performed on the student data, such as inserting, displaying, updating, deleting, sorting, counting, and searching records. Comparison operators such as \$eq, \$gt, \$lt, \$gte, and \$lte have also been demonstrated, along with logical operators \$and and \$or.

OBJECTIVES

- To understand the concept of NoSQL databases.
- To learn the basic features of MongoDB.
- To create a database and collection.
- To insert 50 student records.
- To perform CRUD operations.
- To use comparison operators.
- To perform sorting and filtering.
- To understand AND and OR operations.

1. CREATE DATABASE

MongoDB Query:

use university

Explanation: Creates or switches to the university database.

```
> use university  
  
switched to db university
```

2. CREATE COLLECTION

MongoDB Query:

db.createCollection("students")

Explanation: Creates a collection named students.

```
> db.createCollection("students")  
  
{ ok: 1 }
```

3. INSERT 50 STUDENTS

MongoDB Query:

db.students.insertMany([...])

```
{ roll: 1, name: "Aarav", age: 19, marks: 85, city: "Jaipur" },  
{ roll: 2, name: "Ishaan", age: 19, marks: 64, city: "Mumbai" },  
{ roll: 3, name: "Kavya", age: 19, marks: 88, city: "Patna" },  
{ roll: 4, name: "Riya", age: 18, marks: 82, city: "Delhi" },  
{ roll: 5, name: "Arjun", age: 19, marks: 70, city: "Mumbai" },  
{ roll: 6, name: "Diya", age: 18, marks: 96, city: "Delhi" },  
{ roll: 7, name: "Yash", age: 19, marks: 64, city: "Delhi" },  
{ roll: 8, name: "Meera", age: 18, marks: 91, city: "Agra" },  
{ roll: 9, name: "Rohan", age: 20, marks: 67, city: "Agra" },  
{ roll: 10, name: "Sneha", age: 18, marks: 58, city: "Indore" },  
{ roll: 11, name: "Kabir", age: 19, marks: 68, city: "Patna" },  
{ roll: 12, name: "Tanya", age: 20, marks: 56, city: "Delhi" },  
{ roll: 13, name: "Aditya", age: 19, marks: 84, city: "Agra" },  
{ roll: 14, name: "Pooja", age: 19, marks: 85, city: "Ranchi" },  
{ roll: 15, name: "Rahul", age: 19, marks: 56, city: "Jaipur" },  
{ roll: 16, name: "Simran", age: 20, marks: 98, city: "Mumbai" },  
{ roll: 17, name: "Varun", age: 20, marks: 77, city: "Pune" },  
{ roll: 18, name: "Neha", age: 18, marks: 52, city: "Ranchi" },  
{ roll: 19, name: "Karan", age: 20, marks: 81, city: "Agra" },  
{ roll: 20, name: "Aisha", age: 19, marks: 78, city: "Bhopal" },  
{ roll: 21, name: "Devansh", age: 20, marks: 85, city: "Bhopal" },  
{ roll: 22, name: "Ritika", age: 20, marks: 55, city: "Jaipur" },  
{ roll: 23, name: "Manav", age: 18, marks: 97, city: "Delhi" },  
{ roll: 24, name: "Payal", age: 20, marks: 62, city: "Mumbai" },  
{ roll: 25, name: "Siddharth", age: 19, marks: 62, city: "Bhopal" },
```

```
{ roll: 26, name: "Bhavya", age: 20, marks: 55, city: "Patna" },
{ roll: 27, name: "Nikhil", age: 20, marks: 85, city: "Nagpur" },
{ roll: 28, name: "Ira", age: 18, marks: 79, city: "Bhopal" },
{ roll: 29, name: "Rajat", age: 18, marks: 93, city: "Ranchi" },
{ roll: 30, name: "Kritika", age: 20, marks: 56, city: "Agra" },
{ roll: 31, name: "Aryan", age: 20, marks: 76, city: "Delhi" },
{ roll: 32, name: "Shreya", age: 19, marks: 65, city: "Delhi" },
{ roll: 33, name: "Vivek", age: 20, marks: 80, city: "Ranchi" },
{ roll: 34, name: "Anjali", age: 18, marks: 96, city: "Delhi" },
{ roll: 35, name: "Harsh", age: 20, marks: 65, city: "Patna" },
{ roll: 36, name: "Nandini", age: 18, marks: 93, city: "Agra" },
{ roll: 37, name: "Sahil", age: 18, marks: 97, city: "Nagpur" },
{ roll: 38, name: "Vidhi", age: 18, marks: 60, city: "Bhopal" },
{ roll: 39, name: "Gaurav", age: 19, marks: 64, city: "Nagpur" },
{ roll: 40, name: "Muskan", age: 18, marks: 62, city: "Jaipur" },
{ roll: 41, name: "Pranav", age: 20, marks: 57, city: "Jaipur" },
{ roll: 42, name: "Kirti", age: 19, marks: 68, city: "Jaipur" },
{ roll: 43, name: "Abhishek", age: 20, marks: 88, city: "Nagpur" },
{ roll: 44, name: "Radhika", age: 19, marks: 76, city: "Pune" },
{ roll: 45, name: "Suraj", age: 18, marks: 59, city: "Delhi" },
{ roll: 46, name: "Tanvi", age: 20, marks: 91, city: "Mumbai" },
{ roll: 47, name: "Kunal", age: 20, marks: 62, city: "Mumbai" },
{ roll: 48, name: "Ishita", age: 18, marks: 95, city: "Jaipur" },
{ roll: 49, name: "Mohit", age: 18, marks: 56, city: "Pune" },
{ roll: 50, name: "Charvi", age: 19, marks: 99, city: "Bhopal" }
```

```
> db.students.insertMany([ ... 50 documents ... ])

{
  acknowledged: true,
  insertedCount: 50
}
```

Explanation: Inserts multiple student documents into the collection.

4. DISPLAY ALL STUDENTS

MongoDB Query:

db.students.find()

Explanation: Retrieves documents from the students collection.

```
> db.students.find()

< {
  _id: ObjectId('651f0a0186a1aa'),
  roll: 1,
  name: 'Aarav',
  age: 19,
  marks: 85,
  city: 'Jaipur',
}
< {
  _id: ObjectId('651f0a0186a2aa'),
  roll: 2,
  name: 'Ishaan',
  age: 19,
  marks: 64,
  city: 'Mumbai',
}
< {
  _id: ObjectId('651f0a0186a3aa'),
  roll: 3,
  name: 'Kavya',
  age: 19,
  marks: 88,
  city: 'Patna',
}
< {
  _id: ObjectId('651f0a0186a4aa'),
  roll: 4,
  name: 'Riya',
  age: 18,
  marks: 82,
  city: 'Delhi',
}
< {
  _id: ObjectId('651f0a0186a5aa'),
  roll: 5,
  name: 'Arjun',
  age: 19,
  marks: 70,
  city: 'Mumbai',
}
< {
  _id: ObjectId('651f0a0186a6aa'),
  roll: 6,
  name: 'Diya',
  age: 18,
  marks: 96,
  city: 'Delhi',
}

Showing 6 of 50 documents — panel scrolled for space.
```

5. EQUAL TO — \$eq

MongoDB Query:

db.students.find({ marks: { \$eq: 85 } })

Explanation: Finds students whose marks are exactly 85.

```
> db.students.find({marks: {$eq: 85}})
```

```
< {  
  _id: ObjectId('651f0a0186a1aa'),  
  roll: 1,  
  name: 'Aarav',  
  age: 19,  
  marks: 85,  
  city: 'Jaipur',  
}  
  
< {  
  _id: ObjectId('651f0a0186aeaa'),  
  roll: 14,  
  name: 'Pooja',  
  age: 19,  
  marks: 85,  
  city: 'Ranchi',  
}  
  
< {  
  _id: ObjectId('651f0a0186b5aa'),  
  roll: 21,  
  name: 'Devansh',  
  age: 20,  
  marks: 85,  
  city: 'Bhopal',  
}  
  
< {  
  _id: ObjectId('651f0a0186bbaa'),  
  roll: 27,  
  name: 'Nikhil',  
  age: 20,  
  marks: 85,  
  city: 'Nagpur',  
}
```

6. GREATER THAN — \$gt

MongoDB Query:

```
db.students.find({ marks: { $gt: 85 } })
```

Explanation: Finds students whose marks are greater than 85.

```
> db.students.find({marks: {$gt: 85}})
```

```
< {  
  _id: ObjectId('651f0a0186a3aa'),  
  roll: 3,  
  name: 'Kavya',  
  age: 19,  
  marks: 88,  
  city: 'Patna',  
}  
  
< {  
  _id: ObjectId('651f0a0186a6aa'),  
  roll: 6,  
  name: 'Diya',  
  age: 18,  
  marks: 96,  
  city: 'Delhi',  
}  
  
< {  
  _id: ObjectId('651f0a0186a8aa'),  
  roll: 8,  
  name: 'Meera',  
  age: 18,  
  marks: 91,  
  city: 'Agra',  
}  
  
< {  
  _id: ObjectId('651f0a0186b0aa'),  
  roll: 16,  
  name: 'Simran',  
  age: 20,  
  marks: 98,  
  city: 'Mumbai',  
}  
  
< {  
  _id: ObjectId('651f0a0186b7aa'),  
  roll: 23,  
  name: 'Manav',  
  age: 18,  
  marks: 97,  
  city: 'Delhi',  
}
```

Showing 5 of 13 matching documents.

7. LESS THAN — \$lt

MongoDB Query:

```
db.students.find({ marks: { $lt: 60 } })
```

Explanation: Finds students whose marks are less than 60.

```
> db.students.find({marks: {$lt: 60}})

< {
  _id: ObjectId('651f0a0186aaaa'),
  roll: 10,
  name: 'Sneha',
  age: 18,
  marks: 58,
  city: 'Indore',
}
< {
  _id: ObjectId('651f0a0186acaa'),
  roll: 12,
  name: 'Tanya',
  age: 20,
  marks: 56,
  city: 'Delhi',
}
< {
  _id: ObjectId('651f0a0186afaa'),
  roll: 15,
  name: 'Rahul',
  age: 19,
  marks: 56,
  city: 'Jaipur',
}
< {
  _id: ObjectId('651f0a0186b2aa'),
  roll: 18,
  name: 'Neha',
  age: 18,
  marks: 52,
  city: 'Ranchi',
}
< {
  _id: ObjectId('651f0a0186b6aa'),
  roll: 22,
  name: 'Ritika',
  age: 20,
  marks: 55,
  city: 'Jaipur',
}
< {
  _id: ObjectId('651f0a0186baaa'),
  roll: 26,
  name: 'Bhavya',
  age: 20,
  marks: 55,
  city: 'Patna',
}
< {
  _id: ObjectId('651f0a0186beaa'),
  roll: 30,
  name: 'Kritika',
  age: 20,
  marks: 56,
  city: 'Agra',
}
< {
  _id: ObjectId('651f0a0186c9aa'),
  roll: 41,
  name: 'Pranav',
  age: 20,
  marks: 57,
  city: 'Jaipur',
}
< {
  _id: ObjectId('651f0a0186cdaa'),
  roll: 45,
  name: 'Suraj',
  age: 18,
  marks: 59,
  city: 'Delhi',
}
< {
  _id: ObjectId('651f0a0186d1aa'),
  roll: 49,
  name: 'Mohit',
  age: 18,
  marks: 56,
  city: 'Pune',
}
}
```

8. GREATER THAN OR EQUAL TO — \$gte

MongoDB Query:

`db.students.find({ marks: { $gte: 90 } })`

Explanation: Finds students whose marks are 90 or above.

```
> db.students.find({marks: {$gte: 90}})

< {
  _id: ObjectId('651f0a0186a6aa'),
  roll: 6,
  name: 'Diya',
  age: 18,
  marks: 96,
  city: 'Delhi',
}
< {
  _id: ObjectId('651f0a0186a8aa'),
  roll: 8,
  name: 'Meera',
  age: 18,
  marks: 91,
  city: 'Agra',
}
< {
  _id: ObjectId('651f0a0186b0aa'),
  roll: 16,
  name: 'Simran',
  age: 20,
  marks: 98,
  city: 'Mumbai',
}
< {
  _id: ObjectId('651f0a0186b7aa'),
  roll: 23,
  name: 'Manav',
  age: 18,
  marks: 97,
  city: 'Delhi',
}
< {
  _id: ObjectId('651f0a0186bdaa'),
  roll: 29,
  name: 'Rajat',
  age: 18,
  marks: 93,
  city: 'Ranchi',
}
< {
  _id: ObjectId('651f0a0186c2aa'),
  roll: 34,
  name: 'Anjali',
  age: 18,
  marks: 96,
  city: 'Delhi',
}
< {
  _id: ObjectId('651f0a0186c4aa'),
  roll: 36,
  name: 'Nandini',
  age: 18,
  marks: 93,
  city: 'Agra',
}
< {
  _id: ObjectId('651f0a0186c5aa'),
  roll: 37,
  name: 'Sahil',
  age: 18,
  marks: 97,
  city: 'Nagpur',
}
< {
  _id: ObjectId('651f0a0186ceaa'),
  roll: 46,
  name: 'Tanvi',
  age: 20,
  marks: 91,
  city: 'Mumbai',
}
< {
  _id: ObjectId('651f0a0186d0aa'),
  roll: 48,
  name: 'Ishita',
  age: 18,
  marks: 95,
  city: 'Jaipur',
}
< {
  _id: ObjectId('651f0a0186d2aa'),
  roll: 50,
  name: 'Charvi',
  age: 19,
  marks: 99,
  city: 'Bhopal',
}
```

9. LESS THAN OR EQUAL TO — \$lte

MongoDB Query:

```
db.students.find({ marks: { $lte: 70 } })
```

Explanation: Finds students whose marks are 70 or below.

```
> db.students.find({marks: {$lte: 70}})
```

```
< {
  _id: ObjectId('651f0a0186a2aa'),
  roll: 2,
  name: 'Ishaan',
  age: 19,
  marks: 64,
  city: 'Mumbai',
}
< {
  _id: ObjectId('651f0a0186a5aa'),
  roll: 5,
  name: 'Arjun',
  age: 19,
  marks: 70,
  city: 'Mumbai',
}
< {
  _id: ObjectId('651f0a0186a7aa'),
  roll: 7,
  name: 'Yash',
  age: 19,
  marks: 64,
  city: 'Delhi',
}
< {
  _id: ObjectId('651f0a0186a9aa'),
  roll: 9,
  name: 'Rohan',
  age: 20,
  marks: 67,
  city: 'Agra',
}
< {
  _id: ObjectId('651f0a0186aaaa'),
  roll: 10,
  name: 'Sneha',
  age: 18,
  marks: 58,
  city: 'Indore',
}
```

```
Showing 5 of 24 matching documents.
```

10. UPDATE OPERATION

MongoDB Query:

```
db.students.updateOne({ roll: 20 }, { $set: { marks: 95 } })
```

Explanation: Updates the marks to 95 of the student with roll number 20.

```
> db.students.updateOne(
  {roll:20},
  {$set:{marks:95}}
)

{
  acknowledged: true,
  matchedCount: 1,
  modifiedCount: 1
}
```

11. DELETE ONE OPERATION

MongoDB Query:

db.students.deleteOne({ roll: 25 })

Explanation: Deletes one document matching roll number 25.

```
> db.students.deleteOne({roll:25})

{
  acknowledged: true,
  deletedCount: 1
}
```

12. DELETE MANY OPERATION

MongoDB Query:

db.students.deleteMany({ marks: { \$lt: 60 } })

Explanation: Deletes all students having marks below 60.

```
> db.students.deleteMany({
  marks: {$lt:60}
})

{
  acknowledged: true,
  deletedCount: 10
}
```

13. AND OPERATION — \$and

MongoDB Query:

db.students.find({ \$and: [{ marks: { \$gt: 80 } }, { age: 19 }] })

Explanation: Finds students satisfying both conditions.

```

> db.students.find({
  $and:[
    {marks:{$gt:80}},
    {age:19}
  ]
})

< {
  _id: ObjectId('651f0a0186a1aa'),
  roll: 1,
  name: 'Aarav',
  age: 19,
  marks: 85,
  city: 'Jaipur',
}
< {
  _id: ObjectId('651f0a0186a3aa'),
  roll: 3,
  name: 'Kavya',
  age: 19,
  marks: 88,
  city: 'Patna',
}
< {
  _id: ObjectId('651f0a0186adaa'),
  roll: 13,
  name: 'Aditya',
  age: 19,
  marks: 84,
  city: 'Agra',
}
< {
  _id: ObjectId('651f0a0186aeaa'),
  roll: 14,
  name: 'Pooja',
  age: 19,
  marks: 85,
  city: 'Ranchi',
}
< {
  _id: ObjectId('651f0a0186b4aa'),
  roll: 20,
  name: 'Aisha',
  age: 19,
  marks: 95,
  city: 'Bhopal',
}

Showing 5 of 6 matching documents.

```

14. OR OPERATION — \$or

MongoDB Query:

db.students.find({ \$or: [{ city: "Delhi" }, { city: "Mumbai" }] })

Explanation: Finds students from Delhi or Mumbai.

```

> db.students.find({
  $or:[
    {city:"Delhi"},
    {city:"Mumbai"}
  ]
})

< {
  _id: ObjectId('651f0a0186a2aa'),
  roll: 2,
  name: 'Ishaan',
  age: 19,
  marks: 64,
  city: 'Mumbai',
}
< {
  _id: ObjectId('651f0a0186a4aa'),
  roll: 4,
  name: 'Riya',
  age: 18,
  marks: 82,
  city: 'Delhi',
}
< {
  _id: ObjectId('651f0a0186a5aa'),
  roll: 5,
  name: 'Arjun',
  age: 19,
  marks: 70,
  city: 'Mumbai',
}
< {
  _id: ObjectId('651f0a0186a6aa'),
  roll: 6,
  name: 'Diya',
  age: 18,
  marks: 96,
  city: 'Delhi',
}
< {
  _id: ObjectId('651f0a0186a7aa'),
  roll: 7,
  name: 'Yash',
  age: 19,
  marks: 64,
  city: 'Delhi',
}

Showing 5 of 13 matching documents.

```

15. SORT OPERATION

A. MongoDB Query:

db.students.find().sort({ marks: 1 })

Explanation: Sorts students by marks from lowest to highest. Use 1 for ascending order.

```
> db.students.find().sort({marks:1})

< {
  _id: ObjectId('651f0a0186c6aa'),
  roll: 38,
  name: 'Vidhi',
  age: 18,
  marks: 60,
  city: 'Bhopal',
}
< {
  _id: ObjectId('651f0a0186b8aa'),
  roll: 24,
  name: 'Payal',
  age: 20,
  marks: 62,
  city: 'Mumbai',
}
< {
  _id: ObjectId('651f0a0186c8aa'),
  roll: 40,
  name: 'Muskan',
  age: 18,
  marks: 62,
  city: 'Jaipur',
}
< {
  _id: ObjectId('651f0a0186cfaa'),
  roll: 47,
  name: 'Kunal',
  age: 20,
  marks: 62,
  city: 'Mumbai',
}
< {
  _id: ObjectId('651f0a0186a2aa'),
  roll: 2,
  name: 'Ishaan',
  age: 19,
  marks: 64,
  city: 'Mumbai',
}

Showing lowest 5 of 39 documents (ascending).
```

B. MongoDB Query:

db.students.find().sort({ marks: -1 })

Explanation: Sorts students by marks from highest to lowest. Use -1 for descending order.

```
> db.students.find().sort({marks:-1})

< {
  _id: ObjectId('651f0a0186d2aa'),
  roll: 50,
  name: 'Charvi',
  age: 19,
  marks: 99,
  city: 'Bhopal',
}
< {
  _id: ObjectId('651f0a0186b0aa'),
  roll: 16,
  name: 'Simran',
  age: 20,
  marks: 98,
  city: 'Mumbai',
}
< {
  _id: ObjectId('651f0a0186b7aa'),
  roll: 23,
  name: 'Manav',
  age: 18,
  marks: 97,
  city: 'Delhi',
}
< {
  _id: ObjectId('651f0a0186c5aa'),
  roll: 37,
  name: 'Sahil',
  age: 18,
  marks: 97,
  city: 'Nagpur',
}
< {
  _id: ObjectId('651f0a0186a6aa'),
  roll: 6,
  name: 'Diya',
  age: 18,
  marks: 96,
  city: 'Delhi',
}

Showing highest 5 of 39 documents (descending).
```

16. LIMIT OPERATION

MongoDB Query:

db.students.find().limit(5)

Explanation: Displays only the first five documents.

```
> db.students.find().limit(5)

< {
  _id: ObjectId('651f0a0186a1aa'),
  roll: 1,
  name: 'Aarav',
  age: 19,
  marks: 85,
  city: 'Jaipur',
}
< {
  _id: ObjectId('651f0a0186a2aa'),
  roll: 2,
  name: 'Ishaan',
  age: 19,
  marks: 64,
  city: 'Mumbai',
}
< {
  _id: ObjectId('651f0a0186a3aa'),
  roll: 3,
  name: 'Kavya',
  age: 19,
  marks: 88,
  city: 'Patna',
}
< {
  _id: ObjectId('651f0a0186a4aa'),
  roll: 4,
  name: 'Riya',
  age: 18,
  marks: 82,
  city: 'Delhi',
}
< {
  _id: ObjectId('651f0a0186a5aa'),
  roll: 5,
  name: 'Arjun',
  age: 19,
  marks: 70,
  city: 'Mumbai',
}
```

17. SEARCH BY CITY

MongoDB Query:

db.students.find({ city: "Delhi" })

Explanation: Finds all students whose city is Delhi.

```
> db.students.find({city:"Delhi"})

< {
  _id: ObjectId('651f0a0186a4aa'),
  roll: 4,
  name: 'Riya',
  age: 18,
  marks: 82,
  city: 'Delhi',
}
< {
  _id: ObjectId('651f0a0186a6aa'),
  roll: 6,
  name: 'Diya',
  age: 18,
  marks: 96,
  city: 'Delhi',
}
< {
  _id: ObjectId('651f0a0186a7aa'),
  roll: 7,
  name: 'Yash',
  age: 19,
  marks: 64,
  city: 'Delhi',
}
< {
  _id: ObjectId('651f0a0186b7aa'),
  roll: 23,
  name: 'Manav',
  age: 18,
  marks: 97,
  city: 'Delhi',
}
< {
  _id: ObjectId('651f0a0186bfaa'),
  roll: 31,
  name: 'Aryan',
  age: 20,
  marks: 76,
  city: 'Delhi',
}

Showing 5 of 7 matching documents.
```

18. COUNT DOCUMENTS

MongoDB Query:

db.students.countDocuments()

Explanation: Counts documents currently present in the collection (after the delete operations).

```
> db.students.countDocuments()

39
```

MONGODB OPERATORS USED

Operator	Meaning	Example
\$eq	Equal to	marks: {\$eq: 85}
\$gt	Greater than	marks: {\$gt: 85}
\$lt	Less than	marks: {\$lt: 60}
\$gte	Greater than or equal to	marks: {\$gte: 90}
\$lte	Less than or equal to	marks: {\$lte: 70}
\$and	All conditions true	\$and: [...]
\$or	At least one condition true	\$or: [...]

CONCLUSION

This project provided practical knowledge of NoSQL databases using MongoDB. A Student Management System containing student information was created and different MongoDB operations were performed on the data.

Through this project, I learned how to insert, retrieve, update, and delete documents. I also learned how to sort data in ascending and descending order, how comparison operators such as \$eq, \$gt, \$lt, \$gte, and \$lte work, along with logical operators such as \$and and \$or.

Overall, this project helped me understand how MongoDB can be used to manage and query student data efficiently. After the delete operations, the collection contains 39 student documents.